An Explainable Model for ROP Disease Diagnosis From Multi-centered Infant Fundus Images

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***Abstract*—ROP is an eye disease affecting the retina of a prematurely born infant and is the leading cause of childhood blindness at a global level. Doctors have used pediatric fundus images captured using retina imaging devices to analyze the retina and detect various abnormalities in the infant retina. After conducting a study on various deep learning works related to infant eye disease diagnosis and ROP stage detection, it was observed that the generalizability of models on multi-centered data and the explainability of existing deep learning systems need improvement. This research aims to propose an explainable AI system for diagnosing disease from infant retina images. The proposed study employed multi-centered publicly available datasets related to infant eye images and designed a two-step approach for ROP diagnosis. At first, segmentation is done using a U-net based architecture, then the features are extracted using an Efficientnet architecture, and a support vector machine performs the classification. The proposed study achieved an accuracy of 95.27 percent for ROP detection on multicenter data. The results show that these systems can be used for the clinical diagnosis of ROP, however, the detection of ROP stage is more complicated and requires more attention from the research community.**

***Index Terms*—ROP, ICROP, CAD, SVM, U-Net, Efficientnet**

I. Introduction

An Infant is a baby whose age period is between 0-1 years. Vision starts when she emerges from the womb. If an infant is born at term (GA 37-42 weeks) with mature retinas, the infant initially sees only shadows of nearby objects. Retina development and vision progress rapidly over time, and by the end of one year of age a baby's vision achieves almost the same color vision and depth perception capabilities as those of an adult. During the journey of vision development in the first year infants may be affected by a bunch of diseases such as TORCH (Toxoplasmosis, Other infections, Rubella, Cytomegalovirus, Herpes simplex) infections, congenital anomalies, retinoblastoma, and retinal dystrophies like Leber Congenital amaurosis (LCA), retinoschisis and achromatopsia, Retinal astrocytic hamartomas, Retinal hemorrhages, Optic Nerve Hypoplasia, Toxoplasmosis chorioretinitis infection etc. Performing a proper and prompt diagnosis and treatment can help the baby recover from these diseases. If the baby is born less than 37 weeks of gestational age (GA) and the retina is immature, a classical disease called Retinopathy of Prematurity may affect the retina. Usually, the retina begins to develop during the fourth month in the womb using nutrition, and by the end of 38 weeks, it reaches a mature state. If the baby is born prematurely(around 32 weeks of gestational age), blood vessels stop growing, and abnormal growth of blood vessels occurs. This situation is called ROP. Early detection and treatment are necessary for the disease. The International Classification of Retinopathy of Prematurity(ICROP) [1] defines zones, clinical features of various stages of ROP, Plus diseases, and A-ROP. Practitioners use the ICROP guidelines for the detection of ROP, plus/Preplus, A-ROP, and stage classification. To get an understanding of the zones of the retina, [1, Fig. 1] can be used. Different stages of ROP affected eyes is given in [1].

Depending on which retinal zone the disease has affected, the stages, and the presence or absence of plus disease, it is possible to tell whether a pediatric eye needs treatment or observation. The Early Treatment for Retinopathy of Prematurity [2] defines Type 1 ROP and Type 2 ROP. Type 1 ROP needs urgent treatment, and Type 2 ROP needs observation only. Retinal imaging has enhanced our ability to refine our understanding of the anatomy, pathophysiology, diagnosis, and management of retinal disorders. However,

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neonates and infants have smaller eyes and different retinal anatomy. The characteristics of immature eyes are entirely different from those of mature eyes. There may exist multiple stages in a single ROP image. Viz stage 3 with plus, underdeveloped retina vessels, immaturity of the optic disc and macula, image quality issues like illuminations, external noise, and artifacts present in an image make ROP disease diagnosis, stage classification, and segmentation of interesting regions in the infant retina, a difficult task for both an experienced practitioner and a computer-aided diagnostic system(CAD). The present study proposed an automated CAD system that can be used for ROP detection. The main contributions of this study are the following.

• A Lightweight model is used for faster diagnosis and better utilization of space.

• A Two step approach is used to improve the explainability and generalizability of the system. Two step approach can detect biomarkers such as Blood vessels, Demarcation line or Ridge, and Optic Disc in the images and can explain the reason behind the diagnosis.

• Multi-center infant fundus images were used for model development and testing.

Based on the affected location(zone), stages, and presence of plus disease, it is possible to tell whether a pediatric eye needs treatment or observation. Early treatment for Retinopathy of Prematurity [2] defines Type 1 ROP and Type 2 ROP. If any of conditions i)Zone 1, any stage with plus ii)Zone 1, stage 3 without plus iii)Zone 2, stage 2, or stage 3 with plus is satisfied, then treatment is needed immediately and is called Type 1 ROP. Type 1 ROP is considered as a severe case and treatment is needed. At the same time in the case of Type 2 ROP, only observation is needed. Conditions are, i)Zone 1, stage 1 or 2 without plus or ii)Zone 2, stage 3 without plus. There may exist multiple stages in a single ROP image. Underdeveloped retina vessels and immaturity of the optic disc and macula, and image quality issues like illuminations, external noise, and artifacts present in an image make ROP disease detection, classification, and segmentation a difficult task for both an experienced practitioner and a computer-aided diagnostic system(CAD).

II. Literature survey

Even though various tasks such as ROP detection, stage detection, Plus disease detection, AP-ROP detection, Mild/moderate/severe classification, and multiple infant eye disease detection exist, present study focused on ROP detection. Various CAD systems for the ROP disease detection task are described in the next section. Given an image to a CAD system, the ROP detection aims to detect the presence or absence of ROP in the image. From Table I, it can be concluded that the detection of ROP has achieved a maximum accuracy of 98 percent [3], [4]. However, these models work on single-centered data and do not adequately explain the reason behind the diagnosis. The accuracy of the reviewed papers ranges from 90 to 98. However, single-centered private data are used; hence, a performance comparison based on performance measures is not worthwhile. The explainability of these systems needs to be improved. Future work must concentrate on public multi-centered data, accuracy improvement, and making the system explainable.

III. METHODOLOGY

A. Data Analytics

1) ROP detection: A subset of Dataset 1 from [17] HVDROPDB can be used for the detection of ROP. The dataset contains images taken by using RetCam(Clarity MSI, US) and Neo(Forus Healthcare, Bangalore, India) imaging systems. A total of 85 images were marked as images with ROP stages (35 images from Neo and 50 images from RetCam). 100 images were marked as Normal images (50 images from RetCam and 50 images from Neo). Even though the dataset is well-balanced class-wise, the dataset size is too small.

2) Stage detection: Dataset 2 [18] consists of Normal(240 images) newborn eye images and Stage 1(100), Stage 2(167), Stage 3(265) images of ROP affected eyes. It can be observed that the dataset is imbalanced. The devices used to capture fundus images are RetCam (RetCam, Clarity Medical Systems, Inc. USA), ROP Screening System (SW-8000, China), and Nautilus fundus imaging system (RS-B002, China).

Dataset 3 [19] contains images taken from 3 popular devices: Clarity RetCam 3, Natus RetCam Envision, and Phoenix ICON. The dataset consists of 13 infant eye conditions. It consists of ROP-related infant data classified into Normal, Stage 0, Stage 1, Stage 2, Stage 3, Plus, and A-ROP.

By integrating data from [17], [18], and [19], it is possible to increase dataset size and variability in images. Data augmentation will further increase the size of the data set.

B. Experimental design

The proposed study used data from multiple resources. The dataset design for the ROP detection task using multiple data sources of our study is depicted in the Fig. 1.

C. Proposed system design

The proposed system consists of a segmentation network, a feature extraction network, and a classification network. The segmentation network consists of 3 trained models, each of which performs segmentation tasks on the infant fundus images. BV-SEG, OD-SEG, and RIDGE-SEG perform Blood vessel segmentation, Optic disc segmentation, and Ridge segmentation, respectively. Each network follows an Encoder-Decoder based architecture with Efficientnet ,Madhu et al. [20] as a backbone network for feature extraction. After each segmentation model processes the image, their outputs are combined into a single composite image. This image is then input to the feature extraction model. The study used pre-trained models for feature extraction, and the extracted features from the best layer were given to the classifier for the classification task. The architecture of the system is shown in Fig. 2

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TABLE I: Comparison of ROP detection tasks

Author	Method	Performance	Explainable	Generalizable
Worrall et al. [5]	After Preprocessing and data Auguementation,Googlenet with 2-way softmax used for feature extraction and classification.	0.54-0.72	Yes	Yes
Zhang et al. [3]	Pretrained Alexnet, VGG, and Googlenet were used for feature extraction and classification.	.98	No	No
Wang et al. [6]	Block 1 performs feature extraction using Inception BN architecture. Block 2: Feature fusion block. Block 3: classification done by CNN layers followed by a softmax layer.	.96	No	Yes
Huang et al. [7]	Transfer learning technique is applied using various deep learning models (VGG19, VGG16,MobileNet,Inception V3, DenseNet)	.96	No	No
Attallah et al. [8]	Spatial feature extraction: Transfer learning using Resnet, Inception, Inception-Resnet, and Xception are used. Spectral features were extracted using FWHT. Feature integration phase: Reduced extracted features are integrated using PCA, AE, and DWT. Classification of disease done using L-SVM, Q-SVM, and LDA	0.93	No	No
Wang et al. [9]	An ensemble of Google Inception V3, Xception, InceptionResnet-V2, called CNN group B detects stages of ROP, and does plus/preplus classification	(.89-.96)	Yes	Kappa
Kumar et al. [10]	OD detection done using YOLO-V5. Morphological operations followed by threshold and filtering are used to create a vessel map. OD and vessel maps are used to find Zone and classification.	.88	No	No
Lei et al. [4]	Resnet50 as the base architecture and further improved it by adding a channel and spatial attention module.	.98,.945	Yes	No
Wang et al. [11]	Diabetic retinopathy images are used for training. Then, ROP data is used for testing the model performance	–	No	No
Peng et al. [12]	Multi-semantic feature aggregation (MsFA)	.88(F1) .90	No	Kappa
Sankari et al. [13]	Retinal vessels are segmented using a Modified MultiResUNet, A spatial dependency matrix is created, and contour features extracted. Then, by using permutation importance, the best features are selected. A Random Forest classifier performs classification using these best features	.95	No	No
Sankari et al. [14]	SIFT and SURF features and features from the SegNet,extracted. Resnet, Quantum Support Vector Machine (QSVM) are used as classifiers	0.93-0.96	No	No
Shen et al. [15]	Resnet18, DenseNet121, and EfficientNetB2 for feature extraction.Feature fusion module with a stacking scheme learns the high-level features	.92	No	No
Rao et al. [16]	EfficientNet-B0	.91	No	No

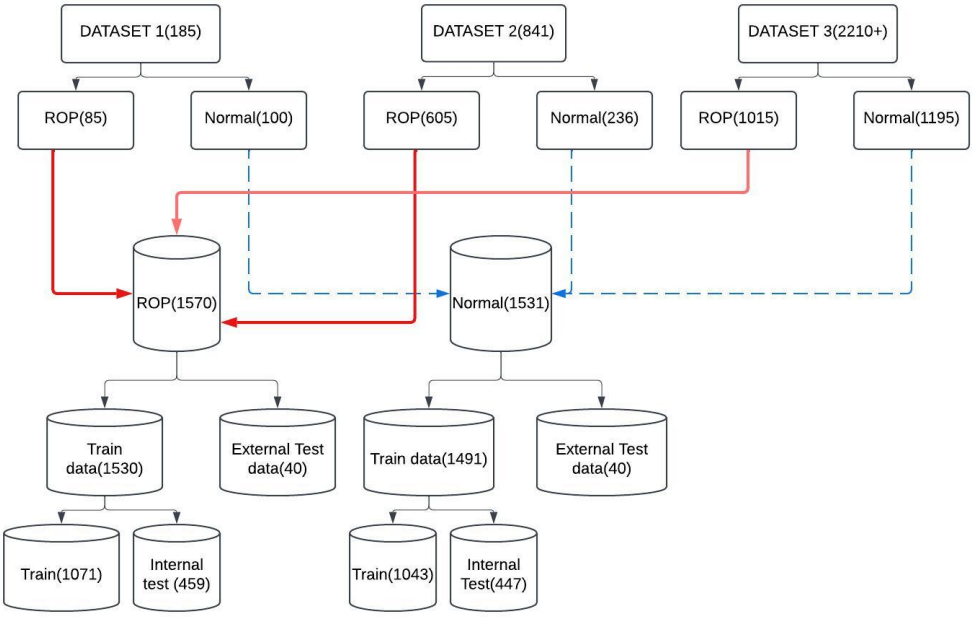


Fig. 1: Data design logic for our ROP detection experiments.

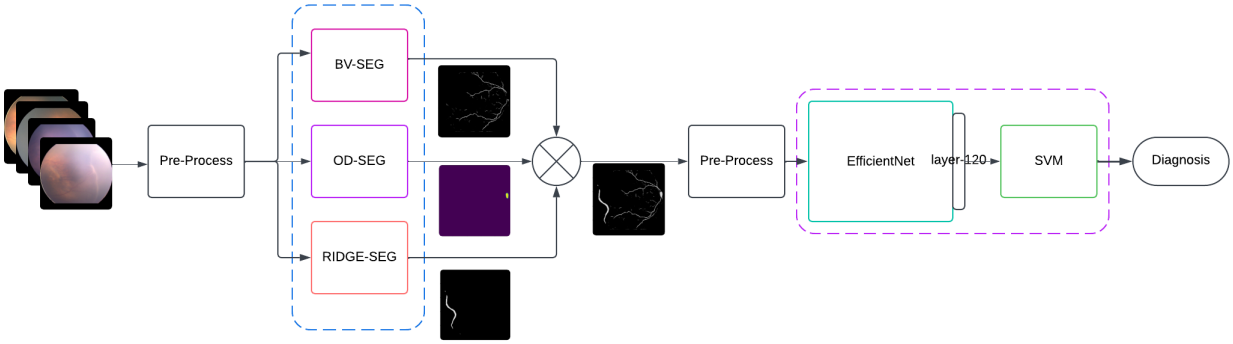


Fig. 2: Proposed Two step approach

1) Preprocessing: For the segmentation task,this study uses a publicly available HVDROPDB [17] dataset, which consists of 300 images. The fundus images of premature babies taken from RetCam and Neo devices. The dataset is divided into 3 subsets and each of the subsets is suitable for the Blood vessel, Optical disc, and ridge segmentation task. Each subset contains 100 original images with the corresponding ground truth. During the pre processing stage, image size of both original and Ground Truth reduced to (128,128,3). Large dimensions will take more training time for the training network. To prevent the loss of information, No cropping is done on the original fundus images. After all of the images were reduced to the standard size , a one to one correspondence between Original and Ground truth image has been made. Then the dataset is divided into 90:10 for training and validation.

2) Segmentation: Then, a U-net model [21] is defined. After ablation studies, a pre-trained model Efficientnetb5 [22] with weights from imagenet [23] was chosen as Unet backbone, Madhu et al. [20]. The backbone extracts features from the data and is used as an encoder of the U-net. The network has been trained using an optimizer algorithm, Adam. Binary cross-entropy loss function was used for this binary semantic segmentation task. Each model is trained for up to 200 epochs with a batch size of 3, each batch consisting of 30 randomly taken images from the training set. The resulting model can segment blood vessels, the optic disc, and the ridges. This segmentation step introduces an explainability to the system.

3) Feature extraction: A Classifier uses the learned features for classification. Deep features from lightweight pre-trained models were employed to automate the feature extraction process. The study preferred lightweight models because they

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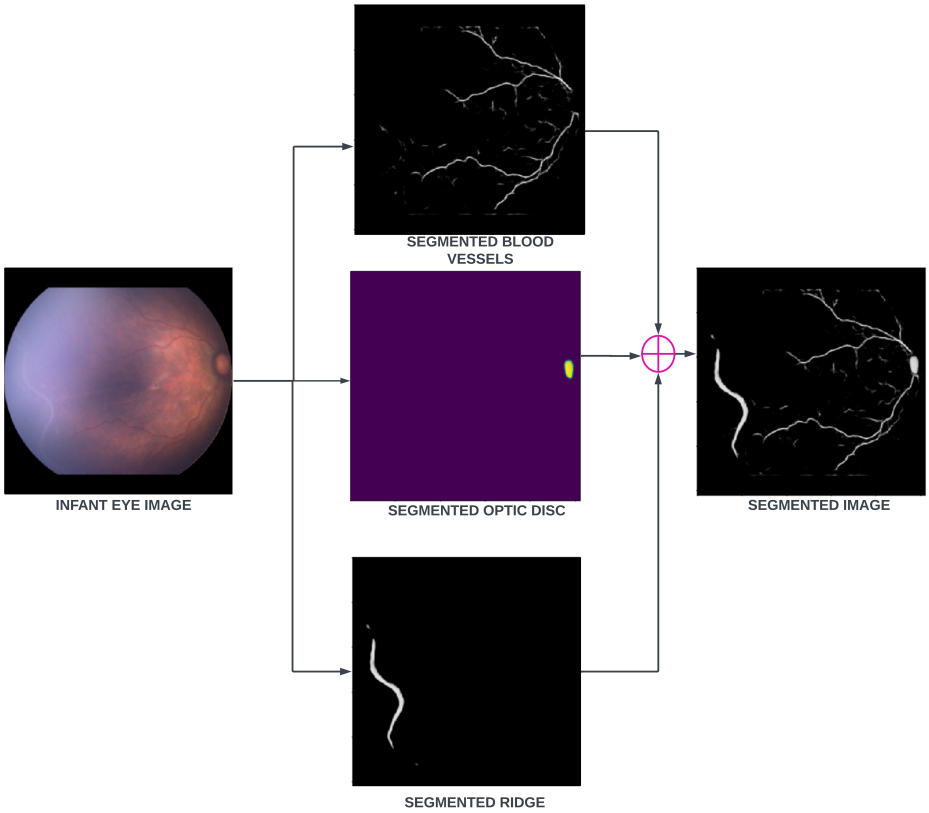


Fig. 3: Segmentation results using the proposed approach. The leftmost image in a row shows the image to be segmented. The middle images show segmented blood vessels, optical disc(OD), and ridge. The rightmost image shows the final segmented image.

produce fewer features and are better in terms of memory space and time utilization. Large pretrained models generate a huge number of features and use more memory space and time. Pre-trained models Effcientnet [24] and Mobilenet [25] were employed for the feature extraction in this study. After performing ablation studies using different pre-trained models with different deep feature layers, it was concluded that Efficentnet with deep layer outperforms other models.

4) Classification: The classification module uses the extracted features and performs the classification task. Following the ablation studies, a support vector machine [26] and deep neural networks are used as classifiers.

IV. RESULTS

The performance of different tasks was evaluated on the test dataset, and the corresponding results are presented in Figure 4. As shown in Fig. 1, internal test data and external test data to test the model. The internal test data set with 906 images shows an accuracy of 95.27 percent. The external dataset consists of 80 images from multi-center data that are not used for training. The system shows an accuracy of 96.25 percent on the external dataset.

A. Comparison of proposed methods and existing methods

The proposed study used multi-centered data from different publicly available datasets related to ROP detection. The system has been compared with [3], [6], [9], [12]. From Table II, it can be concluded that our ROP diagnosis system is more explainable and generalizable than existing systems. The performance of the proposed system is good and can be used for clinical applications.

V. CONCLUSION

Retinopathy of Prematurity (ROP) is one of the most severe eye diseases that affects premature babies. Early detection may help to give prompt and proper treatment. A detailed survey of automated screening systems that use a deep learning-based approach is conducted. Existing systems work on single-centered data, and explainability is low. The study employed multi-centered data and followed a Two-step approach for the detection of ROP and stage classification. The two-step approach adds explainability to the system. Segmentation of various interesting regions of an ROP-affected eye, such as demarcation lines, ridges, and optic disk, was done at first using deep learning models. Then, the feature extraction is done using lightweight models, and the extracted features are given to the classifier. ROP detection shows an accuracy of 95.24

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TABLE II: Comparison of ROP detection tasks

Paper	Feature extraction	Classifier	Accuracy (Internal)	Accuracy (External)	Explainable
Ours	EfficientNet-B0	SVM	95.27	96.25	Yes
Ours	EfficientNet-B0	DNN	95.05	97.50	Yes
Ours	MobileNet	SVM	95.69	98.75	Yes
Ours	MobileNet	DNN	94.62	97.50	Yes
[3]	VGG	VGG	0.98	–	No
[6]	Inception-BN	DNN	0.96	–	No
[9]	Inception-V3, Xception, InceptionResNet-V2	–	0.89–0.96 (Kappa)	–	Moderate
[12]	DenseNet121 (Multi-semantic)	DNN	0.9614	–	No

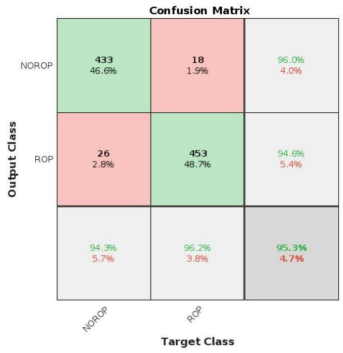


Fig. 4: ROP detection results using multi-centered data shows an accuracy of 95.24.

percent, and on the external dataset, it shows an accuracy of 96 percent. Although ROP diagnosis performance is promising when compared with existing work, future research is needed to improve the accuracy of stage detection. Retinopathy of prematurity detection and stage classification, Grading of ROP, and Multiple eye condition detection from infant's eyes are challenging tasks and need more attention from the research community.

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